Software Requirements Specification

for

ClinicConnect

Version 2.0

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft 2 Version 2.0	Adrienne Jobs Navauldo Sharpe	Second draft created to reflect major revisions to the system's technical environment, including updates to the Operating Environment, Software Interfaces, and Communications Interfaces and Functional Reuirement #6	02/12/2025

1 Overall Description

1.1 Product Context and Need

The product, "ClinicConnect," originates from identified inefficiencies in the current appointment management systems of small to medium-sized medical clinics, as revealed by a recent survey. The survey highlighted that clinics rely on informal, fragmented methods like phone calls, WhatsApp, and social media for booking, leading to challenges with missed appointments, poor cancellation management, and an administrative burden on staff. There is a clear and expressed need from clinics for an affordable digital solution that can centralize scheduling, reduce no-shows through patient reminders, and improve overall operational efficiency. ClinicConnect aims to address this need by providing a streamlined, web-based appointment management system.

1.2 Product Functionality

The major functions of ClinicConnect are:

- To allow patients to view available appointment slots and book new appointments online.
- To enable patients to reschedule or cancel their existing appointments.
- To send automated confirmation and reminder notifications to patients via email or SMS.
- To provide clinic staff with a centralized dashboard to view, manage, and filter all appointments.
- To allow clinic staff to manually add, edit, or cancel appointments to maintain an accurate records.

1.3 Stakeholders and Users Characteristics

- Patients: The primary users. They use the system to book, reschedule, and manage their appointments. Their characteristics vary widely in technical expertise and age. A key goal is to make the system simple and accessible to ensure patient compliance. They are the most important users to satisfy the system's success.
- Clinic Staff (Receptionists): Key secondary users. They are responsible for managing the appointment calendar, viewing daily schedules, and handling appointments made via phone or in-person. They require an intuitive interface to efficiently manage patient flow. Their buy-in is critical for day-to-day operation.
- Doctors/Medical Practitioners: Indirect users. They rely on the accuracy of the schedule managed by the clinic staff but may view their daily appointments. They are less frequent users of the system's administrative functions but are high-priority stakeholders.
- Clinic Administrators/Owners: Important decision-makers. They are concerned with the system's cost, its impact on reducing no-shows, and the overall efficiency gains. They value reporting features and cost-effectiveness.

1.4 Operating Environment

ClinicConnect will be designed as a dynamic, browser-based web application accessible across desktop and mobile devices. The presentation layer will be constructed using standard web technologies (HTML5 and JavaScript) with Bootstrap as the responsive framework, ensuring compatibility with modern web browsers (including current versions of Chrome, Firefox, Safari, and Edge).

The application logic will be implemented in PHP, with a MySQL relational database serving as the persistent data store. The database will house all system entities including appointment records, patient profiles, clinic schedules and user accounts.

For deployment, the application requires a hosting environment supporting the WAMP & MAMP stack (Windows/ MacoS, Apache, MySQL, PHP). Initial deployment will utilize cost-effective hosting solutions that meet these technical specifications.

The minimum technical requirement for end-users is a device with internet connectivity and a web browser released within the last three years.

1.5 Design and Implementation Constraints

- Cost Constraint: The system must utilize exclusively free or open-source technologies throughout its technology stack to address the financial barriers commonly faced by small clinics as impediments to digital utilzation.
- Usability Constraint: User interfaces for both patients, staff and administrative staff must prioritize simplicity and intuitive design, requiring minimal training. The Bootstrap framework will enforce consistent, accessible interface patterns that accommodate varying levels of technical literacy.
- Technology Stack Constraint: Implementation is confined to the PHP, JavaScript, and MySQL technology stack to align with development team competencies and ensure project delivery within established timelines.
- Integration Constraint: Version 2.0 will operate as a standalone system without integration requirements to external Electronic Health Record (EHR) or practice management systems, acknowledging the prohibitive complexity and variability of such integrations.
- Automation Constraint: Initial notification workflows will be staff-initiated rather than fully automated, aligning with existing clinic processes while providing foundational infrastructure for future automation.

1.6 Assumptions and Dependencies

- Connectivity Assumption: Clinics and patients are assumed to have basic, stable internet connectivity for system operation.

- Patient Access Assumption: Patients are assumed to possess either an active email address or mobile phone capable of receiving digital communications.
- Data Integrity Assumption: Clinic staff will manually input appointments created through external channels (for example; telephone bookings) to maintain data integrity within the system.
- System Dependencies: The application's notification functionality is contingent upon successful server configuration and reliable execution of scheduled tasks within the hosting environment.

2 Specific Requirements

2.1 External Interface Requirements

2.1.1 Hardware Interfaces

The primary hardware interface for end-users (patients and staff) is their own device's web browser, which will interact with the system's web server. No special hardware or libraries are required for the core functionality.

2.1.2 Software Interfaces

The server-side application architecture utilizes PHP for business logic implementation with MySQL as the relational database management system. The client interface consists of standards-compliant HTML5 and JavaScript delivered to modern web browsers. External service integrations will communicate through secure HTTP protocols following industry-standard authentication and data exchange patterns.

2.1.3 Communications Interfaces

All client-server communications will utilize HTTPS (TLS 1.2 or higher) as the transport protocol to ensure encrypted data transmission between web browsers and the application server. Third-party service integrations will use secure HTTP requests with appropriate authentication mechanisms. The system will rely on transport-layer security provided by HTTPS rather than implementing custom encryption protocols.

2.2 System Features

Functional Requirement 1

Feature ID: FR-001
Title: Book Appointment Online

Use Case: UC-001 (Patient Books an Appointment)

Rationale: To allow patients to book appointments 24/7 without needing to call the clinic, thereby reducing the administrative burden on staff and providing convenience to patients.

Relates to/Dependencies: None (This is a core, independent feature to start with).

Priority: High

Team Owner: Navauldo Sharpe

User Requirement: The system shall allow a patient to search for available appointment slots and book a new appointment.

System Requirements:

1.1 The system shall display available appointment slots based on the clinic's configured working hours and existing bookings.
1.2 The system shall allow the user to select an available slot and provide necessary details (e.g., patient name, contact information, reason for visit).
1.3 The system shall confirm the booking by generating a unique booking reference and sending a confirmation notification (e.g., email or SMS) to the patient.

Acceptance Criteria:

1. Given a patient is on the booking page, when they select a date, then the system displays only available time slots for that date (i.e., slots not already booked).
2. When a patient submits a completed booking form with valid data, then the system creates the appointment in the centralized record and sends a confirmation message 100% of the time.
3. If a patient attempts to book an already-taken slot, the system shall prevent the booking and display an error message.

Functional Requirement 2

Feature ID: FR-002
Title: Reschedule Appointment

Use Case: UC-002 (Patient Reschedules an Appointment)

Rationale: To provide patients with an easy way to change their appointment time, which can reduce no-show rates by offering flexibility. This addresses the problem of "No way for the patients to reschedule."

Relates to/Dependencies: FR-001 (Online Appointment Booking). Requires an existing appointment to reschedule.

Priority: High

Team Owner: Adrienne Jobs

User Requirement: The system shall allow a patient to reschedule an existing appointment to a new, available time slot.

System Requirements:

2.1 The system shall allow a patient to access their existing appointment using a unique booking reference or registered email.
2.2 The system shall allow the patient to view their current appointment details and see alternative available slots for rescheduling.
2.3 Upon selecting a new slot, the system shall update the appointment record, free up the original time slot, and send a confirmation of the change to the patient.

Acceptance Criteria:

1. Given a patient has a valid booking reference, when they enter it, then the system displays their current appointment details and a button to "Reschedule".
2. When a patient successfully reschedules an appointment, the old time slot becomes available for others to book, and the new time slot is marked as booked.
3. If a patient attempts to reschedule to an unavailable slot, the system shall not permit the change and shall prompt the user to select a different time.

Functional Requirement 3

Feature ID: FR-003
Title: View and Manage Appointments from Dashboard

Use Case: UC-003 (Staff Views and Manages Appointments)

Rationale: To give clinic staff a single, centralized view of all appointments, cancellations, and no-shows, enabling them to track clinic efficiency and manage patient flow effectively.

Relates to/Dependencies: FR-001, FR-002 (The dashboard displays data generated by these features).

Priority: Medium

Team Owner: Navaldo Sharpe

User Requirement: The system shall provide clinic staff with a dashboard to view, filter, and manage all appointments.

System Requirements:

3.1 The system shall display a list of all appointments for a selected date range, showing key details such as patient name, time, status (e.g., booked, cancelled, no-show).
3.2 The system shall allow staff to filter the view by status (e.g., show only "no-shows" for the week).
3.3 The system shall allow authorized staff to manually add, cancel, or edit appointment details to accommodate phone bookings or front-desk changes, ensuring the centralized record is always accurate.

Acceptance Criteria:

1. Upon logging into the staff dashboard, the system displays today's appointments by default.
2. When a staff member applies a filter for "Cancellations" for the current week, the system accurately lists all appointments cancelled within that period.
3. When a staff member manually marks a patient as a "no-show", the system records this status in the centralized record and updates the clinic's no-show statistics in real-time.

Functional Requirement 4

Feature ID: FR-004
Title: Manage Patient Contact Information

Use Case: UC-004 (Staff Updates Patient Details)

Rationale: To allow clinic staff to correct or update patient contact information to ensure reminder notifications and important communications are delivered successfully.

Relates to/Dependencies: FR-001 (Patient data comes from bookings), FR-006 (Accurate data needed for reminders)

Priority: Medium

Team Owner: Jabari McPherson

User Requirement: The system shall allow clinic staff to search for a patient and update their contact information.

System Requirements:

4.1 The system shall provide a search function on the staff dashboard to find patients by name from the patients database table.

4.2 The system shall display the patient's current contact information in an editable form pre-filled with existing data.

4.3 Upon saving changes, the system shall validate the phone number format and update the patient's record in the database.

Acceptance Criteria:

1. Given a staff member searches for "John Smith", when results appear, then the system displays all patients with that name and their current contact info.
2. When a staff member updates a patient's phone number and saves, the system immediately uses the new number for all future SMS reminders for that patient.
3. If a staff member attempts to save an invalid phone number (e.g., letters instead of numbers), the system shall reject the update and display a validation error.

Functional Requirement 5

Feature ID: FR-005
Title: Cancel Appointments

Use Case: UC-005 (Patient Cancels Appointment)

Rationale: To understand why cancellations occur and identify patterns that could help improve clinic operations and reduce future cancellations.

Relates to/Dependencies: FR-001, FR-002

Priority: Medium

Team Owner: Adrienne Jobs

User Requirement: The system shall allow patients to cancel appointments while optionally providing a reason for cancellation.

System Requirements:

5.1 The system shall provide a cancellation interface accessible via the patient's appointment management view.
5.2 The system shall present a list of common cancellation reasons (e.g., "Feeling better," "Scheduling conflict," "Cost concerns") with an "Other" option.
5.3 The system shall free up the appointment slot immediately upon cancellation and store the reason if provided.

Acceptance Criteria:

1. When a patient cancels an appointment, the system makes that time slot available for new bookings within 5 seconds.
2. If a patient selects a cancellation reason, the system stores this data associated with the cancelled appointment.
3. The clinic staff dashboard displays cancellation reasons in appointment history reports.

Functional Requirement 6

Feature ID: FR-006
Title: Schedule and Send Reminders

Use Case: UC-006 (Staff Sends Bulk Reminders)

Rationale: To enable clinic staff to send appointment reminders directly within the application interface, providing patients with timely notifications through their dashboard.

Relates to/Dependencies: FR-001 (Requires booked appointments), FR-003 (Accessed via staff dashboard)

Priority: Medium

Team Owner: Navauldo Sharpe

User Requirement: The system shall allow clinic staff to select multiple upcoming appointments and send in-app reminder notifications that will appear on the corresponding patients' dashboards.

System Requirements:

6.1 The system shall provide a dedicated "Send Reminders" interface within the staff dashboard, displaying appointments scheduled within the next 24-72 hours or with selection capabilities.

6.2 The system shall enable staff to select one or multiple appointments for reminder operations.

6.3 The system shall automatically update each appointment record with a "Recent Reminder Sent" navigation bar and a “Notified” status and timestamp to prevent duplicate notifications.

Acceptance Criteria:

1. Given a staff member selects three appointments scheduled for tomorrow, when they click "Send Actual Notifications to 3 Patients", then all three corresponding patient dashboards shall display new reminder notifications.

2. Given reminder delivery is initiated, the system shall display a confirmation message indicating the count of successfully queued reminders.

3. Given a patient logs into their dashboard, they shall see any pending reminder notifications with clear visual distinction between read and unread status.

Functional Requirement 7

Feature ID: FR-007
Title: View Appointment History

Use Case: UC-007 (Patient Views Their History)

Rationale: To improve the patient experience by giving them a central place to view all their past and upcoming appointments, fostering a sense of control and engagement.

Relates to/Dependencies: FR-004 (Requires a patient to be logged in).

Priority: Medium

Team Owner: Dothaniel

User Requirement: The system shall provide logged-in patients with a profile page showing their upcoming appointments and history.

System Requirements:

7.1 The system shall provide a "My Appointments" page for logged-in patients.

7.2 The system shall display a list of the patient's upcoming appointments with options to cancel or reschedule each one.

7.3 The system shall display a separate list of the patient's past appointments, including the date and status (e.g., completed, no-show, cancelled).

Acceptance Criteria:

1. Given a patient is logged in, when they navigate to "My Appointments," then the system displays all their future appointments.
2. When a patient has both upcoming and past appointments, the system shall display them in two clearly labeled sections.
3. Clicking on a past appointment shall display its details but shall not provide options to cancel or reschedule.

Functional Requirement 8

Feature ID: FR-008
Title: Clinic Schedule Management

Use Case: UC-008 (Staff Manages Clinic Hours)

Rationale: To allow clinic staff to define their working hours and breaks, which forms the foundation for determining available appointment slots for patients.

Relates to/Dependencies: FR-001 (The available slots for booking are derived from this schedule).

Priority: Medium

Team Owner: Navauldo Sharpe

User Requirement: The system shall allow authorized staff to set and update the clinic's default working hours and define closure dates.

System Requirements:

8.1 The system shall provide an interface for staff to set recurring weekly working hours (e.g., Monday-Friday, 9 AM - 5 PM).

8.2 The system shall allow staff to block out specific dates (e.g., public holidays) or time slots (e.g., lunch breaks) as unavailable for bookings.

8.3 The system shall use this configured schedule as the basis for calculating available slots on the public booking page.

Acceptance Criteria:

1. Given a staff member has updated the clinic hours, when a patient views the booking page, then only slots within the new hours are shown as available.
2. When a staff member adds a closure date, the booking page shall show no available slots for that entire day.
3. The system shall prevent the booking of any appointment outside of the configured clinic hours.

Functional Requirement 9

Feature ID: FR-009
Title: Export Appointment Data

Use Case: UC-009 (Staff Exports Appointment Data)

Rationale: To allow clinic staff to export appointment data for basic analysis in spreadsheet programs like Excel.

Relates to/Dependencies: FR-001, FR-003 (Uses appointment data from booking system)

Priority: Low

Team Owner: Adrienne Jobs

User Requirement: The system shall allow staff to export filtered appointment data to CSV format.

System Requirements:

9.1 The system shall provide a simple filter form for selecting date ranges and appointment status.

9.2 The system shall generate a CSV file containing appointment data based on the selected filters.

9.3 The system shall ensure the CSV file is properly formatted for opening in spreadsheet software.

Acceptance Criteria:

1. Given staff select a date range, when they click "Export to CSV", then a file download begins within 3 seconds.
2. The downloaded CSV file opens correctly in Microsoft Excel with proper column headers.
3. The exported data matches the filtered results shown in the staff dashboard.

2.3 Use Case View

2.4 Use Case Narrative

UC-001: Book Appointment Online

Name: Book Appointment Online
Description: Allows patients to search for available appointment slots and book new appointments online without staff intervention.
Primary Actor: Patient
Precondition:

- Clinic schedule is configured in the system
- Patient has internet access to the booking portal
- System is operational and connected to database

Basic Flow:

1. Patient accesses the online booking portal
2. System displays available dates based on clinic schedule and working hours
3. Patient selects desired date
4. System displays available time slots for selected date (excluding booked slots and breaks)
5. Patient selects preferred available time slot
6. System displays booking form with required fields (name, contact, reason for visit)
7. Patient enters valid personal details
8. System validates input data format
9. System creates appointment record with "booked" status
10. System generates unique booking reference
11. System sends confirmation notification (email/SMS) to patient
12. System updates appointment dashboard in real-time

Post-condition:

- Appointment is recorded in centralized system
- Selected time slot is marked as booked
- Patient receives confirmation with booking details
- Staff dashboard reflects new appointment

UC-002: Reschedule Appointment

Name: Reschedule Appointment
Description: Allows patients to change existing appointments to new available time slots.
Primary Actor: Patient
Precondition:

- Patient has an existing confirmed appointment

- Patient has valid booking reference or registered account
- Alternative time slots are available

Basic Flow:

1. Patient accesses appointment management page
2. Patient provides booking reference or logs into account
3. System displays current appointment details and "Reschedule" button
4. Patient selects "Reschedule" option
5. System displays available alternative time slots
6. Patient selects new preferred time slot
7. System validates new slot availability
8. System updates appointment record with new time
9. System frees original time slot for other bookings
10. System sends confirmation of change to patient
11. System updates all relevant records

Post-condition:

- Appointment is moved to new time slot
- Original time slot becomes available for other bookings
- Patient receives update confirmation
- Staff dashboard reflects schedule changes

UC-003: View and Manage Appointments Dashboard

Name: View and Manage Appointments Dashboard
Description: Provides clinic staff with centralized view and management capabilities for all appointments.
Primary Actor: Clinic Staff
Precondition:

- Staff member is authenticated and authorized
- System has existing appointment data
- Staff has appropriate permissions level

Basic Flow:

1. Staff logs into staff dashboard
2. System displays today's appointments by default with key details
3. Staff applies filters (date range, status, provider) as needed
4. System displays filtered appointment list
5. Staff views detailed appointment information
6. Staff performs management actions (edit details, cancel, mark no-show)
7. System validates and processes changes

8. System updates centralized records in real-time
9. Staff monitors appointment flow and clinic efficiency metrics

Post-condition:

- Appointment records are accurately updated
- Dashboard reflects current clinic status
- Changes are logged for audit purposes
- Real-time data is available for reporting

UC-004: Manage Patient Contact Information

Name: Manage Patient Contact Information
Description: Allows clinic staff to search for patients and update their contact information.
Primary Actor: Clinic Staff
Precondition:

- Staff member is authenticated
- Patient database contains existing records
- Staff has search and edit permissions

Basic Flow:

1. Staff navigates to patient management section
2. Staff searches for patient by name using search function
3. System displays matching patients with current contact info
4. Staff selects target patient record
5. System displays patient details in editable form pre-filled with existing data
6. Staff updates contact information (phone, email, address)
7. System validates new data format (especially phone number)
8. Staff saves changes
9. System updates patient record in database
10. System confirms successful update

Post-condition:

- Patient contact information is updated in database
- New information is immediately used for future communications
- Update is logged in system audit trail

UC-005: Cancel Appointments

Name: Cancel Appointments
Description: Allows patients to cancel appointments while capturing cancellation reasons for

analysis.
Primary Actor: Patient
Precondition:

- Patient has an existing confirmed appointment
- Patient has valid booking reference or registered account
- Appointment has not already occurred

Basic Flow:

1. Patient accesses appointment management page
2. Patient provides booking reference or logs into account
3. System displays current appointment details
4. Patient selects "Cancel Appointment" option
5. System displays cancellation reason options (predefined list + "Other")
6. Patient selects reason (optional)
7. System confirms cancellation request
8. System frees the appointment slot within 5 seconds
9. System records cancellation reason if provided
10. System sends cancellation confirmation to patient
11. System updates analytics and reporting data

Post-condition:

- Appointment status changed to "cancelled"
- Time slot becomes available for new bookings
- Cancellation reason stored for reporting
- Patient receives cancellation confirmation
- Staff dashboard and reports reflect cancellation

UC-006: Schedule and Send Reminders

Name: Schedule and Send Reminders
Description: Allows staff to efficiently send batch reminder messages to multiple patients.
Primary Actor: Clinic Staff
Precondition:

- Staff member is authenticated and authorized to manage appointment
- Upcoming appointments exist within the next 24-72 hours
- Associated patient accounts are active and accessible
- Notification subsystem is operational

Basic Flow:

1. Staff member navigates to the "Send Reminders" interface within the staff dashboard
2. System displays a list of appointments scheduled within the next 24-72 hours, each with selection checkboxes and patient details
3. Staff member selects multiple appointments using checkboxes or “Select All Appointments” to send to all patients.
4. System validates that selected appointments are eligible for reminders (future appointments)
5. System generates in-app notifications for each selected patient and stores them in the notification database
6. System updates each appointment record with "Recent Reminders Sent" status, and a reference number
7. System displays confirmation message showing count of successfully queued reminders

Post-condition:

- Selected patients receive reminders in the application
- Appointments are marked in system
- Reminder log is updated with timestamps

UC-007: View Appointment History

Name: View Appointment History
Description: Provides patients with access to their complete appointment history and upcoming appointments.
Primary Actor: Patient
Precondition:

- Patient is registered and logged into the system
- Patient has existing appointment history
- Patient profile is properly linked to appointments

Basic Flow:

1. Patient logs into their account
2. Patient navigates to "My Appointments" page
3. System displays two clearly labeled sections:
    - Upcoming appointments (with cancel/reschedule options)
    - Past appointments (view-only with details)

4. Patient views upcoming appointment details
5. Patient views past appointment history with statuses
6. Patient can cancel or reschedule from upcoming section
7. System updates display after any changes

Post-condition:

- Patient can view complete appointment timeline
- Patient can manage upcoming appointments
- Historical data remains intact and accessible
- Patient has sense of control and engagement

UC-008: Clinic Schedule Management

Name: Clinic Schedule Management
Description: Allows authorized staff to define and manage clinic working hours and closures.
Primary Actor: Clinic Administrator
Precondition:

- Staff member is authenticated with administrative privileges
- System has schedule management module available
- No conflicting appointments exist for schedule changes

Basic Flow:

1. Staff accesses schedule management interface
2. System displays current clinic schedule configuration
3. Staff sets/modifies recurring weekly working hours
4. Staff defines specific closure dates (holidays, etc.)
5. Staff configures daily breaks and unavailable periods
6. System validates schedule conflicts
7. Staff saves new schedule configuration
8. System updates availability calculations
9. System applies new schedule to booking engine

Post-condition:

- Clinic schedule is updated in system
- Booking page reflects new availability rules
- Appointment slots outside new hours are blocked
- Schedule changes are logged for audit

UC-009: Generate Reports

Name: Generate Reports
Description: Enables management to create, view, and export detailed operational analytics reports.
Primary Actor: Clinic Administrator
Precondition:

- User has management-level authentication
- Historical appointment data exists in system
- Reporting period selected has available data
- System has reporting module operational

Basic Flow:

1. Manager accesses analytics and reporting dashboard
2. System displays report generation interface with filter options
3. Manager selects report type, date range, and parameters
4. System validates all input fields and formats
5. System retrieves real-time data from multiple database tables
6. System generates comprehensive report with:
    - Tabular data view
    - Key performance indicators
7. Manager reviews and analyzes report data
8. Manager selects export format (PDF, Excel, CSV)
9. System generates formatted file with proper headers and data integrity
10. System provides download of exported report

Post-condition:

- Accurate report is generated matching live database values
- Export files are created with proper formatting
- Report data is available for decision-making
- All calculations reflect current system state
- Exported files open successfully in external applications

3 Other Non-functional Requirements

3.1 Performance Requirements

- The system shall support up to 100 concurrent users without performance degradation.
- The system shall be available 99.5% of time during clinic operating hours.

3.2 Safety and Security Requirements

Since ClinicConnect will be handling sensitive medical and patient data, system safety and data security are highly important. The platform shall adhere to the Data Protection Act policy to prevent data breaches.

1. All database write CRUD (create, read, update and delete) operations must be ACID (Atomicity, Consistency, Isolation, and Durability.) compliant. In case of transactional failure, rollback procedures shall automatically restore the database to its previous consistent state.This prevents data from being corrupted or lost in the event of unexpected failures.

ClinicConnect shall implement role-based access control to ensure that every user can only access the features, data, and operations appropriate to their designated role within the clinic's operations. Each role will have predefined permissions that determine what actions they can perform and what data they can view or modify.

1. Patients will have limited access to personal information only. They can book, reschedule, and cancel their own appointments, view appointment history, and receive notifications. They cannot access other users' data or internal clinic information.
2. Staff Members (receptionists) will have moderate access to manage appointment scheduling, patient check-ins, and updates to basic patient records, but they will not be able to modify system settings or access financial or management reports.
3. Doctors will have restricted professional access allowing them to view patient records, create clinical notes, and appointment details relevant to their assigned patients or schedules, but not administrative or financial data.
4. Administrators will have full system access, including permissions to create or deactivate user accounts, configure clinic settings, manage all appointments, and generate operational or financial reports.

ClinicConnect shall include data anonymization mechanism (masking) within its reporting and analytics modules to ensure that personal information is not exposed in aggregated datasets or exported reports. When generating analytical summaries or performance reports the system will automatically mask or remove identifiers such as patient names, contact information, medical record numbers, or other unique personal identifiers.

3.3 Software Quality Attributes

ClinicConnect will be developed using a modular, service-oriented design with distinct, independently deployable components for authentication, appointment management and patient records. This architectural approach enables isolated development, testing, and modification of individual modules without impacting system-wide functionality. Consistent adherence to PHP-FIG standards (PSR-1, PSR-2, PSR-4) and well implemented inline documentation will ensure code clarity and facilitate long-term maintainability. A structured version control workflow in Git, coupled with automated testing suites and a continuous integration pipeline, will support controlled, incremental updates with minimal operational disruption.

The system will be designed according to human-centered design principles, minimizing cognitive load and ensuring that primary user tasks such as booking or rescheduling an appointment can be completed in five steps or fewer. Interfaces will be consistently structured with clear visual hierarchies, intuitive navigation patterns, accessible color contrast ratios (WCAG 2.1 AA compliance), and immediate feedback for all user interactions..

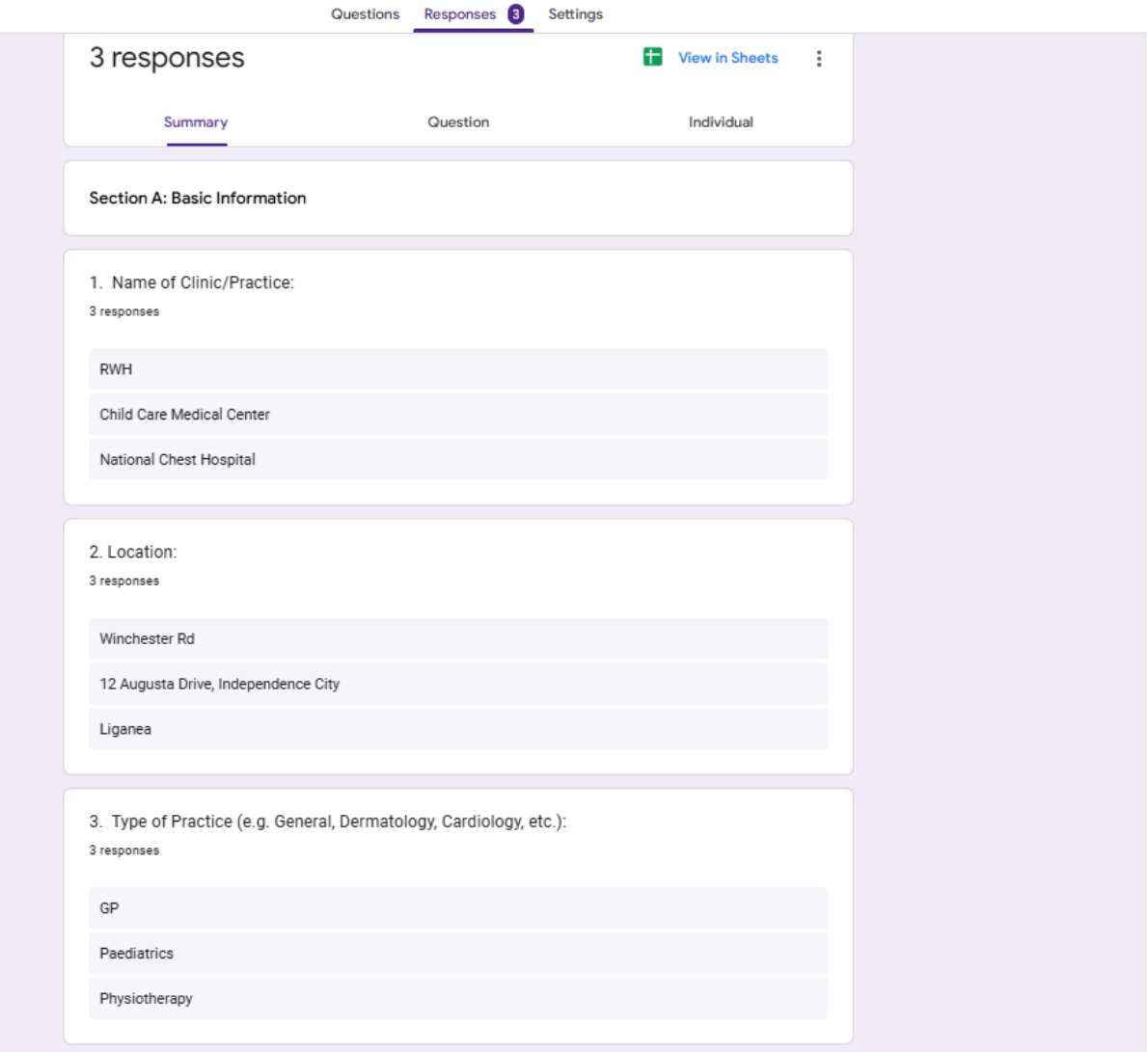
ClinicConnect shall deliver responsive user experiences, with 95% of interactive requests completing within two seconds under standard operational load. Performance optimization will be achieved through efficient database indexing, query optimization, strategic implementation of server-side and client-side caching, and the use of asynchronous processing for non-critical background tasks.

All system interactions will be protected using industry-standard security practices, including encrypted data transmission, secure session management, parameterized database queries to prevent injection attacks, and role-based access control (RBAC) to enforce appropriate data segregation. Regular vulnerability scanning, and adherence to data protection regulations will be integrated into the development lifecycle to safeguard patient information and system integrity.

4 Other Requirements

5 Appendix

As part of the requirements elicitation process, a structured and purpose-driven questionnaire was administered to healthcare professionals and administrative staff across two participating clinics. The primary objective was to gather both quantitative and qualitative insights into existing appointment management practices, operational challenges, and the overall readiness of clinics to transition to a digital based clinic management platform. The questionnaire was conducted between September 18 and September 19, 2025, with participation from Dr. Sherece Green and Professor Minerva Thames, Their feedback provided valuable first-hand perspectives on workflow inefficiencies, and communication gaps between staff and patients. The collected responses were reviewed and analyzed to identify recurring patterns, validate initial user requirements, and refine the functional priorities that ultimately shaped the design of ClinicConnect.



4. Number of doctors/specialists in the practice within your clinic:

3 responses

7

2

3/4

5. Approximate number of patients served per week:

3 responses

25

35

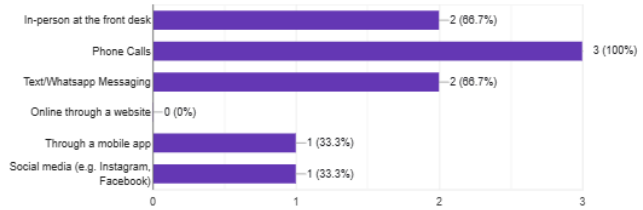
80 plus

Section B: Current Appointment Process

6. How are appointments currently booked? (Select all that apply)

Copy chart

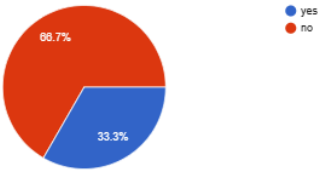
3 responses



7. Do you use any scheduling software or digital tools?

Copy chart

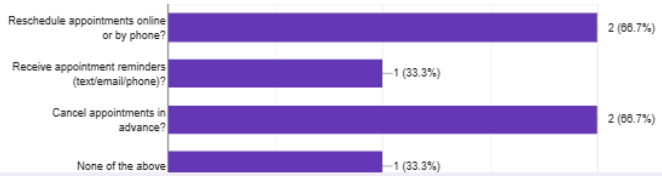
3 responses



8. Are patients able to:

Copy chart

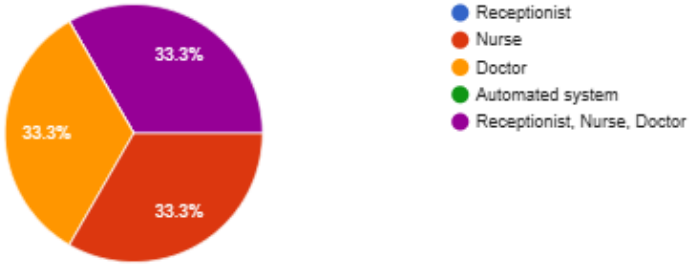
3 responses



9. Who manages the appointment bookings?

Copy chart

3 responses



10. What is the average wait time to get an appointment?

3 responses

No wait for the most part

None

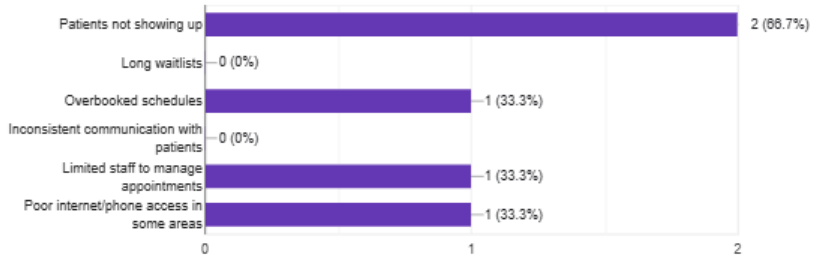
No wait

Section C: Challenges Faced

11. What are the main challenges you face with appointment scheduling? (Select all that apply)

Copy chart

3 responses



12. How have you dealt with missed or cancelled appointments?

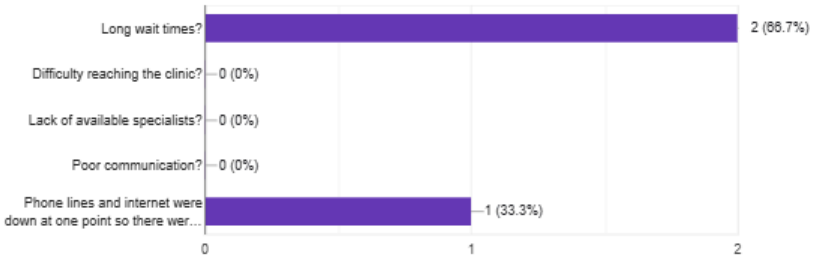
3 responses

- No issues...they come another day
- Just continue to see who comes
- Reschedule

13. Do patients ever complain about:

Copy chart

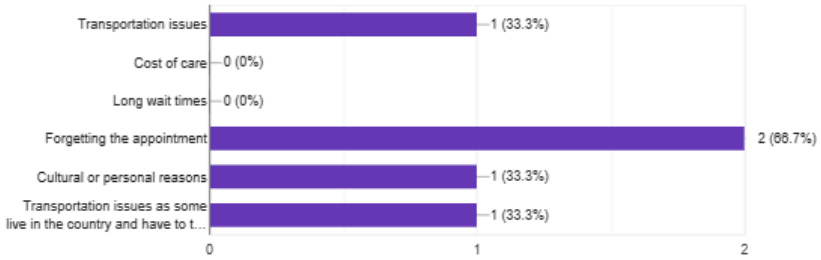
3 responses



14(a). What are some common reasons patients give for missing or delaying appointments?

Copy chart

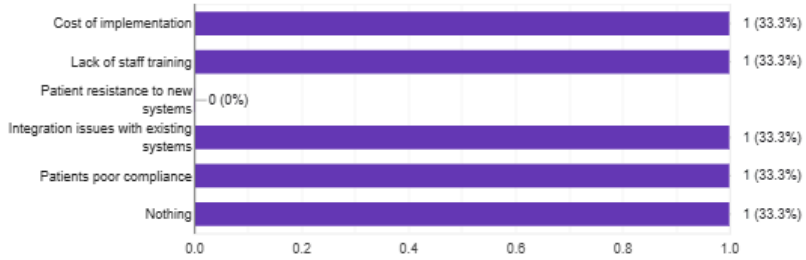
3 responses



14 (b)What is the biggest barrier to adopting new technology for appointment scheduling?

Copy chart

3 responses

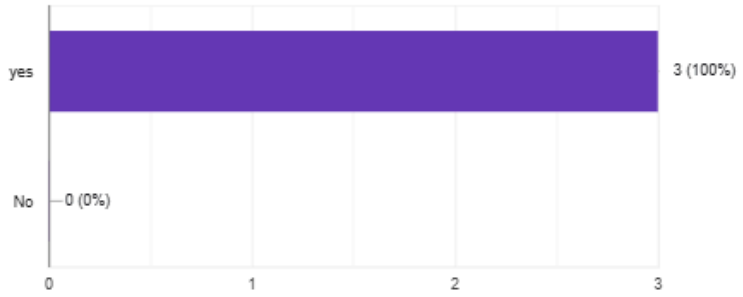


Section D: Technology and Improvement

15 (a). Would you be open to using a digital system to manage bookings and reminders?

Copy chart

3 responses



15 (b) . Explain your reasoning for your answer to question 15 (a) .

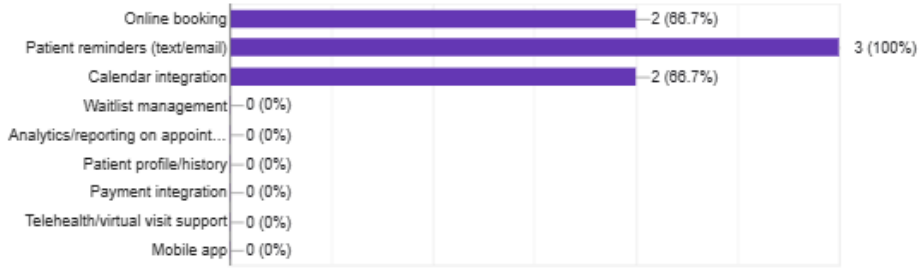
3 responses

- If it's free of cost
- It would improve the service
- No

16. What features would be most useful to you in an appointment management system?
(Select all that apply)

Copy chart

3 responses



Questions Responses 3 Settings

17. Do you think a centralized online booking platform for your clinic (or even across multiple clinics) would benefit your practice and your patients? Why or why not?

3 responses

No

Yes better scheduling

No some ppl are old and cant work theat. Just get ur appointment in person from ur last visit

Section E: Additional Feedback

18. What suggestions do you have for improving the appointment process in your clinic or in Jamaica generally?

3 responses

The clinics and public hospitals in Jamaica are the places that best need a proper appointment system.

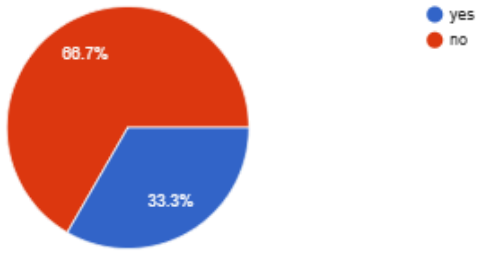
Reminders

Nothing

19. Would you be willing to participate in a pilot test of a new appointment system developed by local software engineers?

Copy chart

3 responses



20. Is there anything else you'd like us to know?

3 responses

No

Poor compliance is a big factor

No

🙏 Thank You!

(Responses)

File Edit View Insert Format Data Tools Extensions Help Ask Gemini

Timestamp	1. Name of Clinic/Practice:	2. Location:	3. Type of Practice (e.g. General, Dermatology, C	4. Number of doctors/specialists in the practice wi	5. Approximate number of patients served p
9/18/2025 16:45:26	RWH	Winchester Rd	GP	7	
9/18/2025 23:14:29	Child Care Medical Center	12 Augusta Drive, Indep	Paediatrics	2	
9/19/2025 11:24:53	National Chest Hospital	Liganea	Physiotherapy	3/4	80 plus

Form Responses 1

(Responses)

	F	G	H	I	J
1	5. Approximate number of patients served per wee	6. How are appointments currently booked? (Selec	7. Do you use any scheduling software or digital tc	8. Are patients able to:	9. Who manages the appointment bookings?
2	25	In-person at the front desk, Phone Calls, Text/Wha	yes	Reschedule appointments onl	Receptionist, Nurse, Doctor
3	35	Phone Calls	no	None of the above	Nurse
4	80 plus	In-person at the front desk, Phone Calls, Text/Wha	no	Reschedule appointments onl	Doctor

Form Responses 1

(Responses)

	K	L	M	N	O
1	10. What is the average wait time to get an appoin	11. What are the main challenges you face with al (Select all that apply)	12. How have you dealt with missed or cancelled a	13. Do patients ever complain about:	14(a). What are some common rea
2	No wait for the most part	Poor internet/phone access in some areas	No issues...they come another day	Phone lines and internet were down at one	Transportation issues, Transportatic
3	None	Patients not showing up	Just continue to see who comes	Long wait times?	Forgetting the appointment
4	No wait	Patients not showing up, Overbooked schedules, l	Reschedule	Long wait times?	Forgetting the appointment, Cultura

Form Responses 1

(Responses)

File Edit View Insert Format Data Tools Extensions Help Ask Gemini

E12

	P	Q	R	S	T
1	14 (b)What is the biggest barrier to adopting new t	15 (a). Would you be open to using a digital syster	15 (b) . Explain your reasoning for your answer to	16. What features would be most useful to you in (Select all that apply)	17. Do you think a centralize
2	Cost of implementation, Lack of staff training, Inte	yes	If it's free of cost	Online booking, Patient reminders (text/email), Cal	No
3	Patients poor compliance	yes	It would improve the service	Patient reminders (text/email), Calendar integratio	Yes better scheduling
4	Nothing	yes	No	Online booking, Patient reminders (text/email)	No some ppl are old and car

Form Responses 1

(Responses)

File Edit View Insert Format Data Tools Extensions Help Ask Gemini

U10

	T	U	V	W	X	Y
1	17. Do you think a centralized online booking platf	18. What suggestions do you have for improving t	19. Would you be willing to participate in a pilot tes	20. Is there anything else you'd like us to know?	@	
2	al No	The clinics and public hospitals in Jamaica are the	no	No		
3	Yes better scheduling	Reminders	yes	Poor compliance is a big factor		
4	No some ppl are old and cant work theat. Just get	Nothing	no	No		

Form Responses 1